

Operating Systems Lab Environment Setup

Ubuntu, WSL, VS Code & Python

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Component	Version / Status
Windows Subsystem for Linux (WSL)	2.5.9.0 (WSL 2), kernel 6.6.87.2
Ubuntu Linux	24.04.3 LTS (noble), Running
Visual Studio Code	1.103.2 (x64), connected via WSL: Ubuntu
Python	3.12.3 (/usr/bin/python3)
Test program	test_env.py executed successfully

Conclusion: All required components (Ubuntu, WSL, VS Code and Python) are installed, configured and verified. The basic test program ran successfully both from the Ubuntu terminal and inside VS Code connected to WSL.

1. WSL Verification

Command: `wsl --version`

```
PS C:\Users\thean> wsl --version
WSL version: 2.5.9.0
Kernel version: 6.6.87.2-1
WSLg version: 1.0.66
MSRDC version: 1.2.6074
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26340.9596
```

WSL version 2.5.9.0 with kernel 6.6.87.2 is installed on Windows (build 10.0.26340.9596).

2. Ubuntu running on WSL 2

Command: `wsl -l -v`

```
PS C:\Users\thean> wsl -l -v
NAME      STATE      VERSION
* Ubuntu  Running    2
PS C:\Users\thean> |
```

Ubuntu distribution is in Running state and uses WSL VERSION 2.

3. Ubuntu and Python Verification

Commands: `lsb_release -a`, `uname -r`, `python3 --version`, `which python3`

```
thean@MSI:/mnt/c/Users/thean$ lsb_release -a
uname -r
python3 --version
which python3
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:       24.04
Codename:      noble
6.6.87.2-microsoft-standard-WSL2
Python 3.12.3
/usr/bin/python3
thean@MSI:/mnt/c/Users/thean$ |
```

Ubuntu 24.04.3 LTS (noble), kernel 6.6.87.2-microsoft-standard-WSL2, Python 3.12.3 at /usr/bin/python3.

4. VS Code Verification

Commands: `code --version`, `cd ~/oslab`, `code .`

```
thean@MSI:/mnt/c/Users/thean$ code --version
1.103.2
6f17636121051a53c88d3e605c491d22af2ba755
x64
thean@MSI:/mnt/c/Users/thean$ cd ~/oslab
thean@MSI:/mnt/c/Users/thean$ code .
thean@MSI:~/oslab$ |
```

VS Code version 1.103.2 (x64) is installed and launched from the WSL terminal.

5. Test Program Creation and Execution

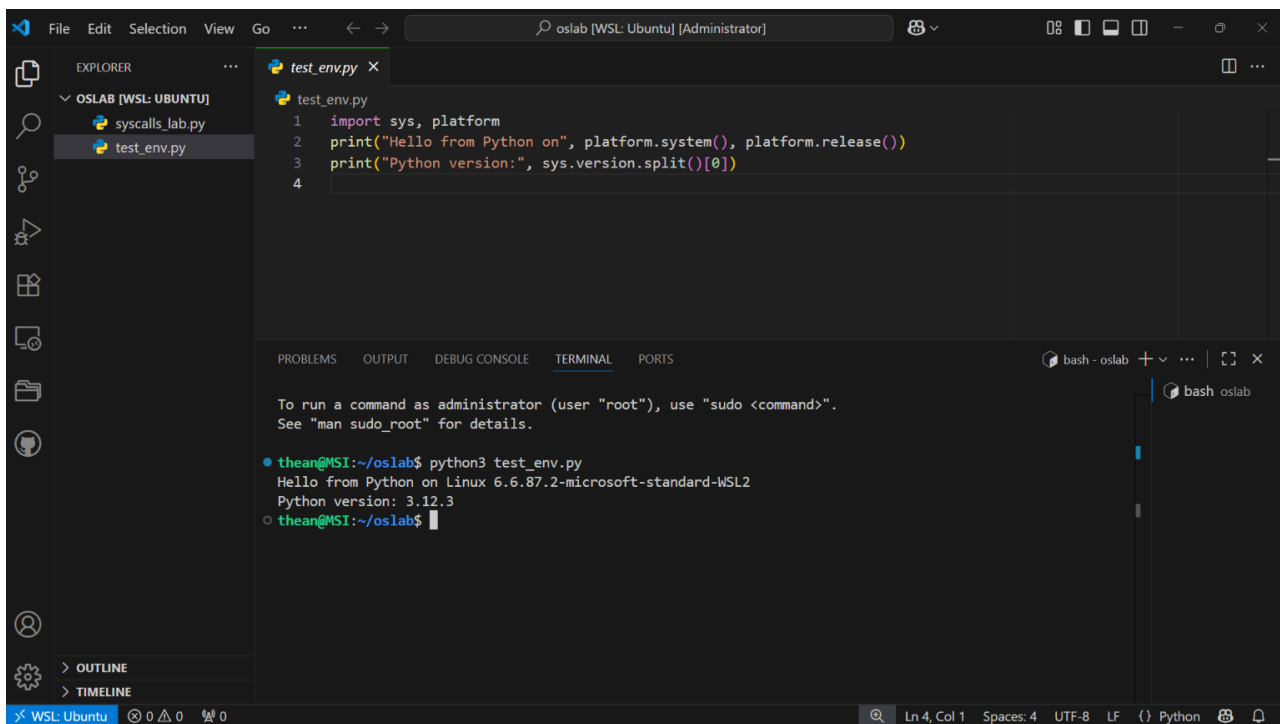
Commands: `echo ... > test_env.py`, `python3 test_env.py`

```
thean@MSI:~/oslab$ cd ~/oslab
echo 'import sys, platform
print("Hello from Python on", platform.system(), platform.release())
print("Python version:", sys.version.split()[0])' > test_env.py
python3 test_env.py
Hello from Python on Linux 6.6.87.2-microsoft-standard-WSL2
Python version: 3.12.3
thean@MSI:~/oslab$
```

Test script was created and executed in WSL; it printed the OS (Linux WSL2) and Python version 3.12.3.

6. Test Program in VS Code (WSL: Ubuntu)

VS Code connected to WSL: Ubuntu, running python3 test_env.py



The screenshot shows the Visual Studio Code (VS Code) interface connected to a Windows Subsystem for Linux (WSL) environment, specifically Ubuntu. The Explorer sidebar on the left shows the file structure with 'test_env.py' selected. The main editor area displays the contents of 'test_env.py', which is a simple Python script. The integrated terminal at the bottom shows the command 'python3 test_env.py' being executed, resulting in the output 'Hello from Python on Linux 6.6.87.2-microsoft-standard-WSL2' and 'Python version: 3.12.3'.

```
test_env.py
1 import sys, platform
2 print("Hello from Python on", platform.system(), platform.release())
3 print("Python version:", sys.version.split()[0])
4
```

```
bash - oslab
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

thean@MSI:~/oslab$ python3 test_env.py
Hello from Python on Linux 6.6.87.2-microsoft-standard-WSL2
Python version: 3.12.3
thean@MSI:~/oslab$
```

VS Code is connected to WSL: Ubuntu (bottom-left). test_env.py ran successfully in the integrated terminal.